Key Performance Indicators for WMO Core Metadata Profile (Version 2)

World Meteorological Organization

Date: 2026-07-07

Version: 2.3.0-draft

Document status: DRAFT

Document location: https://wmo-im.github.io/wcmp2/kpi/wcmp2-kpi-DRAFT.html

Task Team on WIS Metadata (TT-WISMD)[1]

Expert Team on Metadata Standards (ET-Metadata)[2]

Standing Committee on Information Management and Technology (SC-IMT)[3]

Commission for Observation, Infrastructure and Information Systems (INFCOM)[4]

Table of Contents

Chapter 1. Overview

1.1. Purpose

This document is intended to define Key Performance Indicators (KPIs) in support of the WMO Core Metadata Profile (WCMP). KPIs provide measurable and valuable quality assessment rules beyond the rulesets put forth by WCMP.

The core driver of WCMP KPIs is continuous improvement and usability of discovery metadata as part of the WMO Information System (WIS).[5]

1.2. Scope

This document is bound to the WCMP 2 specification[6] and codelists[7]. All other metadata specifications or representations are not in scope.

1.3. Audience

The target stakeholder audiences for this document include (but are not limited to):

- Metadata providers (NCs, DCPCs)
- WIS2 Global Discovery Catalogues (GDCs)
- WIS2 Nodes
- GAW World Data Centres (WDCs)
- WIS2 Monitoring
- Metadata implementers (generation, ingest)

1.4. How to use

The KPIs in this document are designed to help metadata providers curate discovery metadata and for GDCs to measure the quality of metadata from data providers.

To improve quality:

- providers should use the KPIs to build into their metadata generation
- WIS2 Global Services and consumers should use the KPIs to quality assess discovery metadata and provide subsequent feedback to providers

1.5. Scoring

Each KPI assesses a number of criteria associated with metadata quality, resulting in a raw score, as well as a percentage. This approach supports weighted rubric scoring.

1.6. Reference implementation

The TT-WISMD maintains pywcmp[8], as the reference WCMP validation utility, which includes:

- validation against WMO Core Metadata Profile 2, Annex A: Conformance Class Abstract Test Suite (Normative)
- validation against the KPIs described in this document

Documentation on installation, configuration and usage can be found on the pywcmp website.

pywcmp is provided to the community as a resource with continuous improvement. Contributions are welcome and can be facilitated by the WMO.

1.7. Conventions

1.7.1. Symbols and abbreviated terms

Table 1. Symbols and abbreviated terms

Abbreviation	**Term**
AJAX	Asynchronous JavaScript and XML
CSV	Comma-separated values
DCPC	Data Collection and Production Centres
DOI	Digital Object Identifier
GAW	Global Atmospheric Watch
GDC	Global Discovery Catalogue
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
INSPIRE	Infrastructure for Spatial Information in the European Community
JSON	JavaScript Object Notation
MIME	Multipurpose Internet Mail Extensions
NC	National Centre
OGC	Open Geospatial Consortium
pywcmp	WMO implementation of WCMP validation
URL	Uniform Resource Locator
WCMP	WMO Core Metadata Profile
WDC	World Data Centre
WIS	WMO Information System

Abbreviation	Term
WMO	World Meteorological Organization
XHR	XMLHttpRequest

[1] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt/expert-team-metadata-0

[2] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-national-representatives/infcom-management-group/standing-committee-information-management-and-technology-sc-imt/et-metadata

[3] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt

[4] https://community.wmo.int/governance/commission-membership/infcom

[5] https://community.wmo.int/activity-areas/wmo-information-system-wis

[6] https://library.wmo.int/idurl/4/68731

[7] https://codes.wmo.int/wis

[8] https://github.com/wmo-im/pywcmp

Chapter 2. Key performance indicators

2.1. Overview

The table below provides an overview of all KPIs that may be used in the quality assessment of a WCMP record.

Table 2. Table. WCMP Key Performance Indicators

Name	Properties
Good quality title	• `properties.title`
Good quality description	• `properties.description`
Time intervals	• `time.interval` • `additionalElements.temporal.interval`
Graphic overview	• `$.links[?(@.rel=="preview")]`
Links health	• `links[*].href` • `properties.themes[].concepts[].url` • `properties.themes[*].scheme` • `properties.contacts[].links[].href`
Contacts	• `properties.contacts[?(@.roles=="host")] properties.contacts[?(@.roles=="host")].emails properties.contacts[?(@.roles=="host")].contactInstructions properties.contacts[?(@.roles=="publisher")]`
Persistent identifiers	• `properties.externalIds`

2.2. Good quality title

2.2.1. WCMP reference

- 1.7 Properties / Title

2.2.2. WCMP properties

- `properties.title`

2.2.3. Rationale for measurement

The title is the first element of metadata information displayed and helps with initial identification. Meaningful and relevant information makes it easier for users to understand the resource.

In the context of WIS2 Global Discovery Catalogues, the dataset title and description are the two most relevant elements in the WCMP metadata record. These two elements are presented to the users in search results and on the dataset description page. They should highlight the resource's key characteristics to assist users with relevant search results.

2.2.4. Measurement

The length is not too short or too long, contains fewer than three acronyms and is represented in sentence case. Spelling and grammar are correct.

2.2.5. Guidance

The title should be as specific as possible. For example, if the resource contains only one parameter, this can be stated in the title; however, if the resource contains numerous parameters, a more general term should be used in the title, and the parameters should be stated elsewhere in the metadata record (description, themes, keywords, etc.).

2.2.6. Rules

Table 3. Good quality title rules

Rule	Score
The title has 3 words or more	1
The title has 150 characters or fewer	1
The title has only printable characters (numbers and letters) and round brackets	1
The words in title are represented in "Sentence case"	1
The title contains fewer than 3 acronyms (words with all uppercase letters)	1
The title *does not* contain a bulletin header (regular expression: `[A-Z]{4}\d{2}[\s_]*[A-Z]{4}`)	1
The title passes a basic spellcheck	1

Total possible score: 7 **(100%)**

2.3. Good quality description

2.3.1. WCMP reference

- 1.8 Properties / Description

2.3.2. WCMP properties

- `properties.description`

2.3.3. Rationale for measurement

The description facilitates understanding and discovery and is a key element of metadata information displayed in search results. Extensive and meaningful descriptive information enables users to both understand and properly evaluate a metadata record and its respective resource in support of data access, visualization and exploitation.

In the context of WIS2 Global Discovery Catalogues, the product title and description are the two most relevant elements in the WCMP metadata record.

2.3.4. Measurement

The description should neither be too short nor too long and should not contain HTML markup. Spelling and grammar are correct. Bulletin templates should not be used to populate the description.

2.3.5. Guidance

The description should provide a clear and concise statement that enables the reader to understand the content of the dataset.

2.3.5.1. Relevant recommendations

Aim to be understood by non-experts

- Avoid adding a scientific description

Describe the contents of the resource and the key aspects or attributes that are represented

- Limit information in the description to the specific resource that is being described, for example, do not include general background information
- Explain briefly what is unique about this resource and, if appropriate, how it differs from similar resources
- State any other limiting information, such as time period of data validity or geographic coverage

Spell out uncommon acronyms only once

- Avoid jargon and unexplained abbreviations
- Avoid spelling out commonly used acronyms which are already understood by the general public

Write using present or past tense

- Avoid using future verb tense when possible

Use simple paragraph(s) only

- Avoid including HTML and CSV tables or extra spaces to control display of text

Add purpose of data resource where relevant (e.g. for survey data)

- Avoid citing external sources to this resource
- Avoid copying text from a journal article verbatim because this can lead to copyright violation concerns. Additionally, abstracts for journal articles are not intended to describe the provided resource and do not meet the metadata requirements. Related papers can be referenced in the metadata record links.

2.3.5.2. Spell-checking recommendations

- Dictionary by Merriam-Webster: America’s most-trusted online dictionary[1]
- Cambridge Dictionary | English Dictionary, Translations & Thesaurus[2]

2.3.6. Examples

```
"properties": {
    ...
    "description": "Daily climate observations are derived from two sources of data.
The first are Daily Climate Stations producing one or two observations per day of
temperature and precipitation. The second are hourly stations that typically produce
more weather elements, e.g. wind or snow on ground. Only a subset of the total
stations is shown due to size limitations. The criteria for station selection are
listed as below. The priorities for inclusion are as follows: (1) Station is currently
operational, (2) Stations with long periods of record, (3) Stations that are co-
located with the categories above and supplement the period of record.",
    ...
}
```

2.3.7. Rules

Table 4. Good quality description implementation rules

Rule	Score
The description has between 16 and 2048 characters	1
The description does not contain markup (HTML)	1
The description passes a basic spellcheck	1
The description *does not* contain a bulletin template	1

Total possible score: 4 (100%)

2.4. Time intervals

2.4.1. WCMP reference

- 1.11 Geospatial and temporal extents

2.4.2. WCMP properties

- `time.interval`
- `additionalElements.temporal.interval`

2.4.3. Rationale for measurement

Temporal information is a significant characteristic of weather, climate, and water data and is critical for users to know which time periods are covered by datasets and how often new datasets are received.

2.4.4. Measurement

The time interval is present and has a corresponding resolution.

2.4.5. Guidance

A temporal interval and its associated resolution should be expressed for data generated with a given frequency or cadence. For example, ongoing hourly weather observations express a temporal interval with an "open-ended" end time (represented as `..`) and a `resolution` as an ISO 8601 Duration.

2.4.5.1. Examples

```
"time": {
  "interval" : ["2020-10-30", ".."],
  "resolution": "P1D"
}
```

2.4.6. Rules

Table 5. Temporal information implementation rules

Rule	Score
The begin date is before the end date when there is an end date.	1
Only one of the interval extents may be open (begin or end, but not both).	1
For every interval, there is a corresponding resolution present.	1

Total possible score: (begin less than end + only one interval open + resolution) / (total intervals * 3) (100%)

2.5. Graphic overview

2.5.1. WCMP reference

- 1.19 Links and distribution information

2.5.2. WCMP properties

- `$.links[?(@.rel=="preview")]`

2.5.3. Rationale for measurement

A graphic overview of the product provides the user with a high-level preview, which can assist with the initial assessment or evaluation as part of search results presentation.

2.5.4. Measurement

The presence of a `preview` link is checked to ensure that it contains a URL to a common web image file type.[3]

2.5.5. Guidance

In addition to the presence of the graphic overview image it is also valuable to provide consistent image dimensions (e.g. 800x800 pixels) so that all images are normalized and scaling and alignment of overview images can be applied consistently by web applications rendering search results.

Examples of catalogues displaying graphic overviews:

- GISC DWD
- EUMETSAT Product Navigator

2.5.5.1. Example

```
{
  "rel": "preview",
  "type": "image/png",
  "title": "Browse graphic",
  "href": "https://example.org/path/to/browse.png"
}
```

2.5.6. Rules

Table 6. Graphic overview for metadata records implementation rules

Rule	Score
A graphic overview element is present	1
A graphic overview URL resolves successfully	1

Rule	Score
A graphic overview URL content is a common web image file type (check MIME type, content header/magic number)	1

Total possible score: (present link + resolves + image file type) / (total graphic overviews * 3) (100%)

2.6. Links health

2.6.1. WCMP reference

- 1.19 Links and distribution information
- 1.10 Properties / Themes
- 1.12 Properties / Contacts

2.6.2. WCMP properties

All properties with linked information (URLs).

- `links[*].href`
- `properties.themes[*].concepts[*].url`
- `properties.themes[*].scheme`
- `properties.contacts[*].links[*].href`

2.6.3. Rationale for measurement

Broken links damage the user experience and give users the impression that a website is not maintained (88% of online consumers are less likely to return to a site after a bad experience). In addition, numerous broken links can affect the reputation and rank of your website when indexed by mass-market search engines.

HTTPS is increasingly becoming a requirement for numerous agencies as the suggested protocol instead of HTTP. Non-HTTPS links in a WCMP document often lead to mixed content errors in web applications deployed via HTTPS and using AJAX/XHR design patterns. HTTPS supports secure, authoritative and trustworthy links as part of WIS metadata.

2.6.4. Measurement

The number of broken links in each individual metadata record. Broken links include links that, when accessed, result in a 4xx or 5xx HTTP error.[4]

Also being measured is the use of HTTPS (with a valid SSL certificate) as the link protocol throughout WIS metadata.

2.6.5. Guidance

Ensure that all links resolve and are accessible via HTTPS.

2.6.5.1. Examples

```
  "links": [
    {
      "rel": "search",
      "type": "text/html",
      "title": "WOUDC - Data - Station List",
      "href": "https://woudc.org/data/stations"
    }
  ]
```

```
  "links": [
    {
      "rel": "related",
      "type": "application/geo+json",
      "title": "Météo-France, Global Broker Service",
      "href": "mqtts://everyone:everyone@globalbroker.meteo.fr:8883",
      "channel": "cache/a/wis2/#",
      "distribution": {
        "maxMSGsize": 4096,
        "unit": "bytes"
      }
    }
  ]
```

```
  "links": [
    {
      "rel": "subscribe",
      "type": "application/geo+json",
      "title": "Météo-France, Global Broker Service",
      "href": "mqtts://everyone:everyone@globalbroker.meteo.fr:8883",
      "channel": "cache/a/wis2/deu/dwd/data/core/weather/analysis-
prediction/forecast/model/#"
    }
  ]
```

2.6.6. Rules

Table 7. Links health implementation rules

Rule	Score
Link resolves successfully	1

Rule	Score
Link has a valid media type	1

Total possible score: (link resolves) / (total links * 1) (100%)

2.7. Contacts

Metadata records should contain information regarding the contact with the role `host` and how to reach them via email, as well as the role `publisher` for citation purposes.

2.7.1. WCMP reference

- 1.12 Properties / Contacts

2.7.2. WCMP properties

- `properties.contacts[?(@.roles=="host")]`
- `properties.contacts[?(@.roles=="host")].emails`
- `properties.contacts[?(@.roles=="host")].contactInstructions`
- `properties.contacts[?(@.roles=="publisher")]`

2.7.3. Rationale for measurement

Information about the host allows the user to contact the host in case of anything related to accessing the data.

Information about the publisher allows the user to contact the entity that holds, archives, publishes, prints, distributes, releases, issues, or produces the resource.

2.7.4. Measurement

The presence of host information and supporting contact properties (email and contact instructions).

The presence of a publisher role.

2.7.5. Guidance

- Specify a host. Note that a host does not have to be the same as the main point of contact, principal investigator
- Include an email for the host
- Include contact instructions for the host
- Specify a contact with a publisher role
- Use ISO 3166-1 alpha-3 codes to specify the `addresses.country` property.

2.7.5.1. Examples

```
  "properties": {
    ...
    "contacts": [
      {
        "organization": "WMO Lead Centre for Long-Range Forecast Multi-Model
Ensemble",
        "phones": [
          {
            "value": "+82-2-2181-0486",
            "role": "office"
          },
          {
            "value": "+82-2-2181-0489",
            "role": "fax"
          }
        ],
        "emails": [
          {
            "value": "lc_lrfmme@korea.kr",
            "role": "work"
          }
        ],
        "addresses": [
          {
            "deliveryPoint": "61 16-GIL YEOUIDAEBANG-RO DONGJAK-GU SEOUL",
            "city": "SEOUL",
            "postalCode": "07062",
            "country": "KOR"
          }
        ],
        "contactInstructions": "email",
        "links": [
          {
            "href": "https://www.wmolc.org/"
          }
        ],
        "roles": [
          "publisher",
          "host"
        ]
      }
    ]
    ...
  }
```

2.7.6. Rules

By detecting the presence of the contact with the role `host`. See WCMP2, clause 7.

Table 8. Host information implementation rules

Rule	Score
The contact with the role `host` is included.	1
The host contact email is included.	1
The host contact instruction element is included.	1
The contact with the role `publisher` is included.	1

Total possible score: 4 (100%)

2.8. Persistent identifiers

2.8.1. WCMP reference

- 1.14 Properties / Persistent identifiers

2.8.2. WCMP properties

- `properties.externalIds`

2.8.3. Rationale for measurement

Persistent identifiers allow data to be accessible and citable. They make research data easier to access, reuse and verify, thereby making it easier to build on previous work, conduct new research and avoid duplicating already existing work.

2.8.4. Measurement

Whether persistent identifier information is available, can be successfully identified, and provides citation instructions.

2.8.5. Guidance

- Provide a persistent identifier
- Provide a 'Cite as' template as a link object with `rel="cite-as"`

2.8.5.1. Examples

```
"properties": {
```

```
  ...
  "externalIds": [{
    "scheme": "https://doi.org",
    "value": "10.14287/10000001"
  }]
  ...
}
```

For a citation:

```
"properties": {
    ...
    "links": [
    ...
    {
        "rel": "cite-as",
        "title": "Cite as: WMO/GAW Ozone Monitoring Community, World Meteorological
Organization-Global Atmosphere Watch Program (WMO-GAW)/World Ozone and Ultraviolet
Radiation Data Centre (WOUDC) [Data]. Retrieved [YYYY-MM-DD], from https://woudc.org.
A list of all contributors is available on the website. doi:10.14287/10000004",
        "type": "text/html",
        "href": "https://dx.doi.org/10.14287/10000004"
    }]
  ...
}
```

2.8.6. Rules

Table 9. Persistent identifiers implementation rules

Rule	Score
The external identifiers object is present	1
At least one scheme is equal to `https://doi.org`, `https://arks.org`, or `https://handle.net`	1
At least one citation exists as a link object with `rel="cite-as"`	1

Total possible score: 3 (100%)

[1] https://www.merriam-webster.com

[2] https://dictionary.cambridge.org

[3] https://developer.mozilla.org/en-US/docs/Web/Media/Formats/Image_types#Common_image_file_types

[4] https://httpstatuses.com